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A Capsule Network-Based Hybrid Model for Lung Nodule Detection

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Abstract

Lung cancer is a leading cause of death, where early detection is one of the only chances of survival. Manual early detection is time-consuming and error-prone, as it relies on the ability of radiologists to detect small nodules on hundreds of Computed Tomography (CT) images. To address the limitations of human analysis, Computer-Aided Detection (CAD) systems based on Convolutional Neural Networks (CNNs) can assist radiologists but require large datasets, struggle with image variations and lack interpretability. A Capsule Network (CapsNet) can offer an alternative, preserving spatial hierarchies and needing fewer training samples. This work presents a three-stage hybrid model combining You Only Look Once (YOLO) and a CapsNet for lung nodule detection and classification. The pipeline includes YOLOv11 for initial detection, a CapsNet for verification, and YOLOv11 for improved detection. The approach is evaluated on a combined LIDC-IDRI and Lung-PET-CT-Dx dataset. Experiments were done to compare the performance of a Full Image CapsNet classifier versus a Cropped Image CapsNet classifier, where only the Region of Interest (ROI) from detected nodules was used for classification. Results indicate that the Cropped Image CapsNet achieves superior performance with a classification accuracy of 95.57%, compared to 93.07% for the Full Image CapsNet. Despite improved precision, the three-stage pipeline shows slightly lower overall detection accuracy (89.98%) than standalone YOLOv11 (90.79%) due to increased False Negatives (FNs). While a CapsNet enhances classification reliability for FNs, further improvements in bounding box generation and a CapsNet architecture are needed to mitigate its higher False Positive (FP) rate. Future research should optimize the CapsNet, explore alternative detection models, and apply multi-class classification to distinguish benign from malignant nodules. The findings of this study contribute to the ongoing efforts in improving automated lung cancer diagnosis, offering a approach that leverages both CNN-based object detection and a CapsNet's spatial feature encoding capabilities to enhance nodule detection accuracy and reduce radiologist workload.

Keywords: Lung Cancer Detection; Capsule Network; You Only Look Once; Nodule Detection; Nodule Classification; Hybrid Model

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1. Introduction

Lung cancer is one of the most common types of cancer and also one of the deadliest [9, 45, 42]. The later lung cancer is detected, the higher the mortality rate [45, 17]. Early signs of lung cancer are difficult to detect as there are no clear symptoms [32]. Only small nodules in the lung, often between 10 mm and 30 mm in diameter, can be malignant and indicate lung cancer [32, 20, 52]. To identify these nodules, a CT scan is one of the most common performed methods [43]. Detecting these nodules is a challenging and time-consuming task to check all layers to find nodules as their size, appearance and location vary [32, 31, 36]. To avoid missing malignant nodules, in particular, CAD can be integrated into the diagnostic process [46, 13, 20]. The development of CAD systems for lung cancer diagnosis usually involves two central tasks: the detection of lung structures and the classification of lung nodules [46, 32]. One of the leading and most established methods in CAD for medical imaging are Deep Learning (DL) techniques, in particular CNNs [1, 25, 49, 48, 10]. CNNs have established themselves as a state of the art method as they have proven their superiority in numerous studies and achieved better results to date [1, 28, 10]. However, a very large dataset is required to fully exploit these positive features of CNNs, which is not that common for a clinical patient group [2, 1, 10, 50]. In addition, CNNs have weaknesses that make them difficult to use universally. First and foremost is the inability to learn and understand changing perspectives of an image [10, 26, 48]. For example, CNNs have difficulty recognizing images that are mirrored, shifted or rotated [8, 10, 1]. A reason for this is that they use pooling operations to downsample the feature maps and enhance the model's robustness [35].

To address central weaknesses of CNNs, in particular the need for very large datasets and the insufficient consideration of spatial relationships, the CapsNet was developed by Sabour et al. [35, 11, 1, 39]. The architecture of a CapsNet focuses on the relative position between the overall object and its parts [35, 11, 37]. In contrast to CNNs, a CapsNet does not use pooling layers, but encodes properties and positions directly within the architecture [16, 47]. By avoiding pooling operations, the dimensionality of the data is retained and spatial positions and distances can be better learned [1, 48, 16, 47]. As a result, the model requires less data to generalize and is also more robust against noise and unexpected properties [35, 37, 18, 10].

Research on the use of a CapsNet for object detection is still in its early stages, with some approaches such as DeformCaps [27] and detail-oriented models tested [37] on specific datasets but not yet applied to lung nodule CT images [27, 37, 53]. Two-stage detectors often combine a CapsNet with other models such as YOLO to take advantage of a CapsNet for specific classification tasks [51, 24].

In regards to the above mentioned current medical challenges and the advances in the fields of classification and object detection, this work proposes a novel approach for object detection based on a YOLO and a CapsNet. The aim is to combine the strengths of a CapsNet and object detection techniques of YOLO to detect and classify lung nodules on CT and PET/CT images of the lung. This approach should help radiologists and other specialists to gain an initial overview of CT images and to focus their attention on relevant areas of the usually large amount of data available for a patient. This work aims to address the mentioned research gap by developing an approach to localize lung nodules based on CT images using a CapsNet. To overcome the limitations of the CapsNet originally designed for classification, a combination of a CapsNet and a YOLOv11 is used. This approach is then compared with other current methods of object detection.

Building on this foundation of research gap and relevance, the following research questions are formulated to guide the study:

1. How effective is a CapsNet trained on ROI-based cropped nodule images compared to a CapsNet trained on full CT and PET/CT images?
2. Can the integration of a CapsNet support or improve the performance of a YOLOv11 model in object detection for the detection of nodules in CT and PET/CT images?

2. Theoretical Background

To establish the foundation and comprehension of the proposed hybrid model, this section provides the theoretical background of YOLO and CapsNet. This exploration clarifies the strengths and limitations of each approach, highlighting their role in enhancing lung nodule detection. Developed in 2016, YOLO is a one-stage object detection system that has been continuously developed over the years. Today, it is considered as a state of the art method for (real-time) object detection [23, 6].

One of the latest YOLO architectures is YOLOv11, developed by Ultralytics. This builds on the foundations of YOLOv8 and significantly extends and improves its functions [23, 22, 21]. The YOLOv11 architecture basically consists of three main components: the backbone, the neck and the head [23, 6, 21]. The model uses Spatial Pyramid Pooling Fast (SPPF) and Cross Stage Partial with Spatial Attention (C2PSA) in the backbone for feature extraction, and combines these with a multi-stage process in the neck to improve the processing of features at different scales. These different feature maps are passed to three detection heads to detect at different scales, which enables a robust object detection and classification at varying sizes [23, 6, 21, 3]. This modular structure enables YOLOv11 to accurately recognize objects in real time while providing high efficiency and scalability [21, 23].

The weaknesses of CNNs, such as lack of robustness, the need for large datasets and the problem of pooling, are addressed by Sabour et al. through the introduction of a CapsNet, which focuses specifically on spatial dependencies [14, 39, 16]. The characteristics of a CapsNet are the capsule layers, because unlike in a fully connected layer, in a capsule layer the neurons are separated into capsules [14, 16]. Thus, a capsule is a wrapper that groups several neurons [14, 16]. If the capabilities and output of a neuron are compared with the capabilities and output of a capsule, the differences become clear. The single neuron of a fully connected layer, for example, calculates a scalar value from several input values that represents the probability of the presence of a feature [14, 16]. A capsule can also do this, here the group of neurons is used for that and the probability of the presence of a feature can be calculated by the length of the output vector of this capsule [14, 39]. Since the individual neurons no longer calculate the probability but only the vector length, they can focus on properties such as location, skew, etc. [14]. A basic CapsNet initially

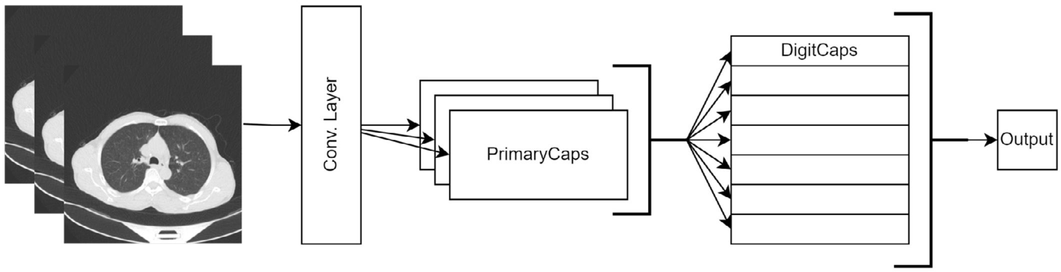

Fig. 1: Simplified overview of a CapsNet architecture. Adapted from [39].

consists of a classic convolution layer with an activation function, whose task is to convert the image into a feature map (see Figure 1) [39, 14, 41, 16]. This feature map is then passed to the second layer, the so-called PrimaryCaps, which has the goal of converting the feature map with the scalar values into vectors [39, 14, 41, 16]. For this purpose, the PrimaryCaps consist of several channels, each with a grid of capsules (see Figure 1) [39, 14]. As a CapsNet does not use pooling, a method must be used to extract the most important features for the next layer of capsules. The dynamic routing, also known as routing by agreement, is used for this [39, 15]. By using routing by agreement, it is checked how well the predictions from the lower capsules, which are PrimaryCaps in this case, match the actual outputs of the upper capsules, the DigitCaps in this example [41, 2]. If a prediction from a lower capsule matches well with the activation of a DigitCaps capsule, the coupling coefficient for this connection between the lower capsule and the DigitCaps capsule is increased, which means that more information is passed from the lower capsule to the DigitCaps capsule [14, 39, 15]. The output is then passed to the DigitCaps layer, this layer only consists of n capsules, where n stands for the number of possible classes [39, 14, 41, 16]. These two layers are fully connected and previously mentioned routing by agreement ensures that each capsule receives the entire input of the primary output [39, 14, 41]. These n capsules now use their vectors to calculate the probability for their respective classes [39, 14].

3. Methodology

In this work, a hybrid three-stage detection and classification model pipeline is proposed in which individual models (YOLOv11 and CapsNet) utilize their detection and classification strengths to support another YOLOv11 model to provide more robust and better results than a standalone YOLOv11. The pipeline consists of three stages: first, candidate regions are detected using YOLOv11. In the second stage, a CapsNet classifies these detected regions as nodules or non-nodules. Finally, the verified nodules are consolidated for visualization and evaluation (see Figure 2). The first stage model is a YOLOv11 model which is trained with a balanced dataset of CT lung images with and

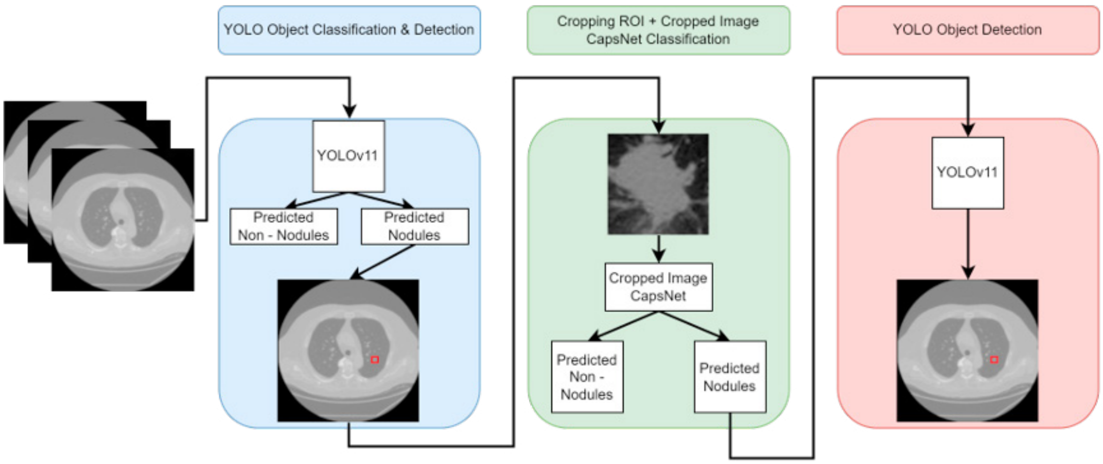

Fig. 2: Proposed YOLO-CapsNet Hybrid Model.

without nodules. A YOLOv11 is the currrent state of the art YOLO model of Ultralytics [22]. YOLOv11 is available in various model sizes, from Nano (n), Small (s), Medium (m) and Large (l) to Extra-Large (x) [21, 22]. These models differ in their scaling, the number of parameters and thus also in their performance [21, 22]. In the proposed pipeline, the model in size m is used. This model has the task of scanning the input image and outputting bounding boxes representing suspected nodule locations. These suspected nodules are cropped as ROIs, based on the bounding boxes, scaled to 128×128 pixels and sent to the second stage. The CapsNet is based on a binary classification to distinguish whether a nodule is present or not. To apply this classification, the suggested bounding box of the first stage is used as ROI from the original image to crop the image to the potential nodule and then pass it to the CapsNet. If the image is classified as a nodule, the original image of the cropped image is sent to the third stage. In the final stage, a YOLOv11 is used, which is the same model as in the first stage. It is also trained with a balanced dataset of CT lung images with and without nodules, and during testing and production, it only receives full images whose ROIs were previously classified as nodules by the CapsNet. This step provides the final output of the pipeline and ensures that only verified nodules are presented for further analysis or clinical decision making. The CapsNet is trained using a learning rate of 0.0005 and a batch size of 16. Early stopping is employed with a patience parameter of 5 epochs, applied to both full-resolution and cropped image inputs. In contrast, the YOLOv11 models were trained using default hyperparameter settings, which included a learning rate of 0.01, a batch size of 16, and early stopping with a patience of 100 epochs. For this work, a combination of two datasets are considered:

1. *The Lung Image Database Consortium* (LIDC) and *Image Database Resource Initiative* (IDRI), called LIDC-IDRI [4].
2. *A Large-Scale CT and PET/CT Dataset for Lung Cancer Diagnosis*, called Lung-PET-CT-Dx [29].

The LIDC-IDRI dataset includes 1,010 thorax CT cases with annotations from four radiologists. In a two-stage review, the radiologists first independently labeled lesions in three categories: *nodule* $\geq 3mm$, *nodule* $< 3mm$, and $non - nodule \geq 3mm$. In addition to the patients with annotations, there are also patients who could not be assigned to any of these three categories, as no nodules or similar could be found here and they can be classified as healthy in this context. However, this dataset does not differentiate between benign and malignant nodules , meaning that the nodules can represent cancerous tumors, scar tissue, or other benign findings. Therefore only those patients from this dataset that have no nodules in the entire CT image are used [4]. The Lung-PET-CT-Dx dataset consists of CT and PET-CT images of 355 lung cancer patients [29]. The images were retrospectively acquired from patients with suspected lung cancer who had undergone PET/CT, among other procedures. Patients were grouped according to cancer diagnosis, adenocarcinoma, small cell carcinoma, large cell carcinoma and squamous cell carcinoma [29]. All CT and PET/CT images from the Lung-PET-CT-Dx dataset are considered. This results in a combined dataset of 53,688 CT and PET/CT images, after manual preprocessing, which is used for training and testing the models.

The performance of a CapsNet trained on cropped images only is compared to a CapsNet trained on nodule and non-nodule images. To analyze the classification performance of the Cropped Image CapsNet in combination with a YOLOv11 , these models are compared with standalone versions, considering only correct classifications. To evaluate the effectiveness of the overall system, it is compared to a standalone YOLOv11 model to see if the combination of specialized models can improve performance.

4. Results

In this Section, the results of the individual models as well as the proposed approach of this work are presented. To evaluate the performance of a CapsNet, the Cropped Image CapsNet is first compared with the Full Image CapsNet. Subsequently, the Cropped Image CapsNet is compared with the partial approach of the proposed model YOLOv11 + Cropped Image CapsNet and a standalone YOLOv11. Finally, the entire proposed model is compared with a standalone YOLOv11.

4.1. Comparison of Cropped Image CapsNet vs Full Image CapsNet

Table 1: Results of the CapsNet models on full and cropped images.

Metric	Full Image CapsNet	Cropped Image CapsNet
Accuracy	93.07%	**95.57%**
Precision	92.36%	**96.42%**
Recall	94.00%	**94.67%**
F1-Score	93.17%	**95.53%**

The training lasted 71 epochs (53 h total training time) for the Full Image CapsNet, with a validation accuracy of 92.94%. The Cropped Image CapsNet, on the other hand, only trained for 29 epochs (20 h total training time) until early stopping was triggered, with a validation accuracy of 95.45% for the best model. One reason for the longer training duration for the full images is that they contain significantly more features. These additional features must be recognized and trained by the model, which leads to a longer training process due to the larger number and the associated higher complexity.

The tests show that the Cropped Image CapsNet correctly categorizes 5,161 out of 5,400 images, while the Full Image CapsNet correctly categorizes 4,998 out of 5,370 images. In the performance metrics, this is reflected in a higher accuracy of 95.57% for the Cropped Image CapsNet compared to 93.07% for the Full Image CapsNet, as well as improved precision values of 96.42% compared to 92.36% and a F1-Score of 95.53% compared to 93.17% (see Table 1).

4.2. Comparison of YOLOv11 + Cropped Image CapsNet vs Cropped Image CapsNet vs YOLOv11

For the evaluation of the combination of a YOLOv11 and a Cropped Image CapsNet, the test dataset is sent once directly to the Cropped Image CapsNet in a cropped version, once as full images to the YOLOv11 and also once first to the YOLOv11 model, after which images were cropped using the suggested bounding boxes and then forwarded to the Cropped Image CapsNet.

Regarding classification, the standalone YOLOv11 model achieved the best performance with an accuracy of 98.16%, correctly classifying 5,271 out of 5,370 images, with only 66 FPs and 33 FNs (see Table 2). This is also reflected in a high recall of 98.78% and a F1-Score of 98.18%. The combined model of a YOLOv11 and a Cropped Image CapsNet achieves an accuracy of 96.55% by correctly categorizing 5,185 out of 5,370 images. It has the lowest

Table 2: Results of classification task between Cropped Image CapsNet, YOLOv11+Cropped Image CapsNet and YOLOv11.

Metric	Cropped Image CapsNet	YOLOv11+Cropped Image CapsNet	YOLOv11
Accuracy	95.57%	96.55%	**98.16%**
Precision	96.42%	**97.87%**	97.59%
Recall	94.67%	95.22%	**98.78%**
F1-Score	95.53%	96.53%	**98.18%**

number of FPs (56), but at the expense of a higher FN count (129), and achieves the highest accuracy of 97.87%. The standalone Cropped Image CapsNet performs the weakest compared to the other models with an accuracy of 95.57%, a precision of 96.87%, a recall of 94.67% and a F1-Score of 95.53%.

4.3. Comparison of Proposed Model and YOLOv11

Table 3: Results of the detection task between the proposed model and YOLOv11.

Metric	Proposed Model	YOLOv11
Accuracy	89.98%	**90.79%**
Precision	**90.17%**	89.51%
Recall	88.97%	**91.67%**
F1-Score	89.57%	**90.58%**
AP50	**95.68%**	95.54%
AP75	**63.67%**	63.33%

In this Subsection, the results of the proposed model, consisting of YOLOv11 + Cropped Image CapsNet + YOLOv11, are compared with the results of the standalone YOLOv11 model.

The proposed model achieves an accuracy of 89.98% by correctly recognizing 2,395 objects (True Positives (TP)) and correctly classifying 2,616 images without objects (True Negatives (TN)) (see Table 3). It has 297 unrecognized nodules (FN) and 261 incorrectly recognized nodules or insufficient Intersection over Union (IoU) with at least 0.5 (FP). The standalone YOLOv11 model achieves a higher accuracy of 90.79%, with 2,475 correctly recognized objects (TP) and 2,604 correctly classified images without objects (TN), as well as 290 poorly recognized nodules (FP) and 225 undetected nodules (FN). Although the standalone YOLOv11 performs better in terms of accuracy (90.79%), recall (91.67%) and F1-Score (90.58%), the combined model shows a better precision of 90.17% and a slight improvement in Average Precision 50 (AP50) (95.69%) and Average Precision 75 (AP75) (63.67%).

To classify the increased number of FN, the distribution of the tumor types adenocarcinoma, small cell carcinoma, large cell carcinoma and squamous cell carcinoma was taken into account. For this purpose, the test set was compared with the FN cases. This showed that out of a total of 20,894 adenocarcinoma tumors (66.20%), 211 cases were classified as FN, which corresponds to a proportion of 71.04% and thus represents an increase of 4.84 percentage points (see Table 4). A similar picture emerges for large cell carcinoma, where 3 out of 201 test images were classified as FN representing an increase of 0.37 percentage points (see Table 4). In contrast, there are 23 FN cases among 3,116 test images for small cell carcinoma, which corresponds to a decrease of 2.13 percentage points. For squamous cell carcinoma, 60 FNs are detected in 7,351 images, which corresponds to a reduction of 3.09 percentage points (see Table 4).

Table 4: Analysis of FN distribution based on tumor types for the proposed model.

	Adenocarcinoma	Small Cell Carcinoma	Large Cell Carcinoma	Squamous Cell Carcinoma
test dataset	66.20%	**9.87%**	0.64%	**23.29%**
FN	**71.04%**	7.74%	**1.01%**	20.20%

5. Discussion

In the first comparison, a CapsNet trained on full CT and PET/CT images is compared to a CapsNet trained on cropped images only. The results show that the model trained on cropped images performs better in all performance metrics, including accuracy, precision, recall and F1-Score (see Table 5). One explanation for these differences is that cropping the images specifically targets the relevant features of the potential lung nodules. This reduces the probability that the model takes irrelevant areas of the images into account. This assumption is supported by the training processes of the two models, as the Cropped Image CapsNet could be trained significantly more efficiently, which is reflected in both a lower training loss in the epochs and an overall shorter training duration. These positive characteristics regarding computational efficiency and the lower number of epochs required for good results have also been noted for the hybrid approach of a YOLO and a CapsNet for tomato plant diseases [24]. With an accuracy of 96.55 %, the combined model achieved similarly good classification results as the tomato disease classification with 97.00% [24]. However, other hybrid models that, for example, combine YOLO with a Tumor–Node–Metastasis (TNM) classifier to classify tumors can deliver better results with 98% accuracy [46]. The results support answering the research question of whether it is beneficial to analyze only the ROI for a CapsNet rather than the entire image. These findings suggest that focusing on ROI does indeed improve performance. However, it is important to note that these results were obtained under idealized conditions. In real-world applications, where a model such as a YOLO automatically determines the ROI, inaccuracies and variability in image quality could affect performance and cause results to deviate from the idealized test conditions.

Table 5: Overview of all results.

	Cropped Image CapsNet vs Full Image CapsNet		**YOLOv11 + Cropped Image CapsNet vs Cropped Image CapsNet vs YOLOv11**			**Proposed Model vs YOLOv11**	
	Cropped Image CapsNet	**Full Image CapsNet**	**YOLOv11 + Cropped Image CapsNet**	**Cropped Image CapsNet**	**YOLOv11**	**Proposed Model**	**YOLOv11**
Accuracy	**95.57%**	93.07%	96.55%	95.57%	**98.16%**	89.98%	**90.79%**
Precision	**96.42%**	92.36%	**97.87%**	96.42%	97.59%	**90.17%**	89.51%
Recall	**94.67%**	94.00%	95.22%	94.67%	**98.78%**	88.97%	**91.67%**
F1-Score	**95.53%**	93.17%	96.53%	95.53%	**98.18%**	89.57%	**90.58%**
AP50	-	-	-	-	-	**95.68%**	95.54%
AP75	-	-	-	-	-	**63.67%**	63.33%

The aim of the second comparison is to evaluate the practicability of the proposed approach in terms of classification performance and to what extent the performance of a Cropped Image CapsNet changes when the ROIs are no longer manually optimized but given by an automated detection model such as a YOLOv11. The results show that the combination of a YOLOv11 and a Cropped Image CapsNet provides better overall performance than a Cropped Image CapsNet alone, especially by reducing FPs and improving accuracy (see Table 5). A simultaneous comparison with a standalone YOLOv11 model also confirms that CapsNet offers added value regarding FPs reduction. Although a YOLOv11 alone achieves a high detection/classification rate, it does so at the cost of an increased number of FPs and therefore a lower precision value than the combined model. The combination with a CapsNet helps to reduce these misclassifications at the cost of increasing number of FNs which result in worse results for the other metrics if compared to the standalone YOLOv11.

In the final analysis, the actual detection ability of the models is compared. The results show that the standalone YOLOv11 model performs better overall in terms of detection, which is supported by higher accuracy, recall and F1-Score values (see Table 5). In direct comparison with other CapsNet architectures, the proposed model shows a lower performance. For example, the DECAPS model achieves an object detection accuracy of up to 94.02% [36]. However, as this model focuses on the detection of other pathologies in CT images, a direct performance comparison is of limited value [36]. Nevertheless, the proposed model shows better precision compared to the YOLO model, meaning that it produces fewer FPs. At the cost of a overly tolerant model on nodules, the Cropped Image CapsNet often misclassifies images of lung nodules as non-nodules, which are called FN. These FNs are particularly problematic in the medical field as they can lead to delayed diagnosis and treatment, which is especially critical in the diagnosis of cancer [7]. This problem has already been noted in literature reviews, precision is an important measure, but only on the condition that the recall is also high to ensure that no tumors are overlooked [33]. This could not be achieved with these results. One possible cause for the occurrence of FNs could be the inaccuracy of the bounding boxes generated by a YOLOv11, which do not always accurately represent the actual ROI. This could lead to a CapsNet not working with optimal image sections and thus negatively influence its decision making. Other reasons may be the type of cancer, as both adenocarcinoma and large cell carcinoma are increasingly declared FN (see Table 4). In contrast to small call carcinoma and squamous cell carcinoma, both types of cancer tend to be found peripherally, which means that the proposed model may not detect them well despite the majority of adenocarcinoma images [34, 12, 40]. The low FP value is also reflected in the Average Precision (AP), where the proposed model achieves slightly better results. However, these results should be interpreted with caution, as the FNs produced by the Cropped Image CapsNet and the first YOLOv11 cannot be considered in the AP, since the Cropped Image CapsNet lacks a confidence value. Therefore, considering only the first YOLOv11 without the CapsNet would present an incomplete picture. The second research question, whether a CapsNet can support or improve the performance of a YOLOv11, is answered with both positive and negative findings. A CapsNet, especially the Cropped Image CapsNet, can support the performance of a YOLOv11 in terms of FP rate. However, an overall performance improvement is not observed, as the overall performance of the system is affected by an increased FN rate. On the other side, one important aspect remains, the proposed model continues to show better precision, meaning that it produces fewer FPs. Looking at the overall performance, this low FPs number suggests on the other side that the Cropped Image CapsNet is critical in classification and therefore often misclassifies images of lung nodules as non-nodules (FN), thus not giving the YOLOv11 model a chance to reverify them. This increased number of FNs is an important problem, especially in the medical sector and particularly for cancer diagnosis, as it is preferable for doctors and patients to diagnose one more potential cancer candidate than to overlook a nodule because of delayed diagnosis and treatment [7]. The causes of these shortcomings may lie in a suboptimal trained CapsNet or in inaccurate bounding boxes used to crop the images. The low input resolution of 128×128 pixels of the CapsNet represents a potential restriction due to computational limitations, as small or hard to recognize features may not be adequately detected. Peripheral tumors in particular are less well detected, indicating insufficient feature extraction. A promising approach for improvement could be the use of patch-based inputs to increase the effective resolution and potentially detect peripheral tumors more specifically [5]. These challenges emphasize the need for a more precise alignment between ROI detection and classification for real improvement and potential use in lung cancer detection. This has also been tackled by others through approaches to improve the YOLO model using other transfer learning models, different loss functions, etc., which could lead to improved ROIs [30, 19, 44]. Furthermore, replacing dynamic routing with other routing approaches could lead to improved performance or results despite poor ROIs [38, 36, 27].

6. Conclusion & Outlook

In this work, a novel hybrid three-stage model for object detection of lung nodules in CT and PET/CT images is presented. In the first step, the images are processed with a YOLOv11 to generate initial bounding boxes. The extracted areas are then classified using a CapsNet. Finally, the areas classified as nodules are analyzed again with a YOLOv11 in the original image to enable more precise detection. This approach enables radiologists to carry out the time-consuming process of searching for potential tumors in just a few seconds. By comparing the Full Image CapsNet and the Cropped Image CapsNet, it is found that cropped images provide better results by focusing on relevant features. In the comparisons between the proposed model and a standalone YOLOv11, it can then be determined that although a YOLOv11 still delivers better results overall, the proposed model can at least provide better precision with the downside of an increased FN rate. It can therefore be said that a CapsNet in the form of a Cropped Image CapsNet as part of a three-stage detector cannot yet support or improve a YOLOv11 in the object detection of nodules in lung CT and PET/CT images.

The implementation of this work highlights several limitations of both the model and the work itself. The CapsNet, in particular, faces challenges due to its lack of a pooling layer, leading to high computational costs. This requires training with a maximum image size of 128×128 pixels due to memory and computing power constraints, even when using a NVIDIA GeForce RTX 3070 GPU with 24 GB VRAM. Increasing the image resolution would have caused significant memory bottlenecks and longer computation times. These resource limitations also resulted in a model with 8 capsules, resembling a basic CapsNet. It is likely that more optimal architectures could be developed with different numbers of capsule layers and varying capsule counts per layer at the price of significantly higher computer resource requirements or longer running times. Additionally, this work does not provide a general conclusion about hybrid approaches with a CapsNet, as it was only examined in conjunction with a YOLOv11 for lung cancer detection. The suitability of this hybrid three-stage approach for other tasks, application areas and architectures remains to be investigated.

This work introduces a three-stage model that opens up new research avenues. Future research could expand the classification strategy from binary to multiclass or hierarchical, enabling differentiation between types of nodules. Optimizing the CapsNet architecture could enhance performance and efficiency, allowing for higher resolution image processing. For this purpose, the approaches for improving the CapsNet and YOLO models mentioned in the discussion could be applied. Additionally, exploring newer YOLO models or alternatives like Faster R-CNN could further boost detection and classification accuracy. These improvements could enhance the model's ability to classify nodules accurately and efficiently, especially in medical applications like lung cancer detection.

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